

Liquid Blending in Pharmaceutical Industries: Ethanol & Water

Project Report

CHE460 - Design & Characterization of Mixing in Pharmaceutical Industries

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Abstract

This report presents the design and characterization of a liquid blending system for preparing a 20% v/v ethanol-water co-solvent mixture for the pharmaceutical application of oral and nasal sprays. A comprehensive review of mixing theory, fluid properties, and equipment options was carried out to select an efficient, sanitary, and scalable configuration suitable for GMP environments. The final design consists of a 2.0 m³ flat-bottom, vertical cylindrical tank constructed from 316L stainless steel, equipped with four baffles and a single 45° four-blade pitched-blade turbine. Key geometric ratios were selected within recommended design ranges ($D/T = 1/3$, $H/T = 1.0$, $C/T = 1/3$), and hydrodynamic performance was evaluated using dimensionless numbers. At an impeller speed of 3.83 s^{-1} , the mixer operates in a fully turbulent regime ($Re \approx 5.3 \times 10^5$) with a power number of 1.27, Froude number of 0.69, and pumping number of 0.79. The corresponding power input is approximately 1.5 kW, or 0.75 kW/m³, which lies within the recommended range for vigorous liquid-liquid blending. Using standard mixing correlations, the time to reach 99% homogeneity is predicted to be about 25 s, with an associated energy input of ~38 kJ and a negligible temperature rise, justifying the isothermal assumption. Overall, the proposed system provides rapid, energy-efficient, and hygienic blending suitable for pharmaceutical co-solvent preparation.

Chapter 1. Introduction

Liquid blending is a critical operation in the pharmaceutical industry. It is used to achieve homogeneity between two or more miscible liquids that can dissolve in each other, ensuring uniformity in product composition and quality. This process plays an important role in the production of solutions, syrups, disinfectants, paints, cosmetics, food products, and various liquid formulations. Miscible liquid blending is essential in the pharmaceutical industry for various purposes such as the uniform distribution of active pharmaceutical ingredients (APIs), preparing sterile solutions, disinfectant products, etc. (Global Center for Pharmaceutical Industry, 2024).

The miscibility of liquids depends on molecular interactions such as polarity and hydrogen bonding. Miscibility tables are commonly used to determine solvent compatibility in pharmaceutical and chemical manufacturing (Cole-Parmer, n.d.). For example, water and ethanol are completely miscible due to their ability to form hydrogen bonds and their polarity compatibility. These tables help determine whether solvents can dissolve uniformly in one another and guide appropriate solvent selection for formulation processes.

Several factors affect the blending process including impeller type, rotational speed, fluid viscosity, and tank geometry. Proper blending ensures that every batch meets the pharmaceutical manufacturing standards by maintaining uniformity in composition, and compliance with quality specifications. (Tamburini & Cipollina, 2020).

The objective of this project is to design a mixing system capable of achieving complete molecular-level homogeneity between miscible liquids and have completely dissolved within each other. The blending of miscible liquids is usually done by convective mixing, driven by the bulk fluid motion generated from an impeller, followed by molecular diffusion that eliminates remaining concentration gradients. (Paul, Atiemo-Obeng, & Kresta, 2004).

Although the general blending process is similar across systems, it is important to find suitable parameters for different liquids, as they may vary in their physical and chemical properties. Parameters such as Reynolds number are crucial in characterizing flow behaviour. Turbulent flow ($Re > 10,000$) is generally preferred for rapid mixing. Additionally, selecting appropriate impeller types, such as propellers, Rushton turbines or pitched-blade turbines, is essential to ensure proper flow patterns and shear conditions within the tank. Additionally, the viscosity and density of liquids must be considered because they directly influence power consumption and mixing performance.

Optimizing these parameters is essential to enhance performance while reducing both time and energy consumption. Poor blending can lead to variations in solvent strength, inconsistent API solubility, batch variability and regulatory non-compliance. Mixing time often acts as a bottleneck factor in the process, meaning that the production cannot proceed until mixing is complete. Reducing mixing time increases overall output, allowing pharmaceutical firms to improve productivity and profitability (Paul et al., 2004). Through the evaluation of fluid characteristics, equipment selection, and operating conditions, this project aims to design an effective liquid blending system that meets pharmaceutical requirements.

Chapter 2. Comprehensive Review of Mixing Theory

2.1 Mixing Theory and Background

Effective mixing is fundamental to the successful design and operation of stirred tank reactors used for the blending of miscible liquids such as water and ethanol. This chapter provides a comprehensive review of the physical properties of the fluids involved, the theoretical basis of mixing processes, and the hydrodynamic and operational considerations associated with impeller-driven blending. Understanding these background concepts is essential for the design of a mixing system capable of achieving rapid homogenization within isothermal batch conditions. The subsections that follow outline the chemical and physical characteristics of the working fluids, the principles governing miscible liquid mixing, and the fluid dynamic mechanisms responsible for convective and diffusive blending.

2.1.1 Defining the Mixing System and Application

This mixing system will be designed for liquid blending for the pharmaceutical industry. Liquid blending is defined as two or more liquids that completely dissolve with one another and are able to reach homogeneity on a molecular level. In the pharmaceuticals industry, miscible liquid blending is used in many products such as injectables, ophthalmic solutions, and mouthwash, and many others. (Paul Mueller Company, 2018) With about 40% of approved drugs and almost 90% of drugs that are candidates to be approved for the market being poorly water-soluble drugs, one prominent example of liquid blending of miscible liquids is preparing

co-solvent. (Xie et al., 2024) Since many active pharmaceutical ingredients (API) and excipients do not dissolve well in pure water alone, water is mixed with other organic liquids such as ethanol, propylene glycol, glycerol, etc must be mixed to create a solvent that increases the solubility of other excipients and API. (Xie et al., 2024)

Water and ethanol mixtures are commonly used in the pharmaceutical industry as cosolvents and helps with preservation (Parke & Birch, 1999). Cosolvents such as water and ethanol create hydroalcoholic solutions that increase the solubility when pharmaceutical products have some ingredients that are more alcohol soluble and others that are more water soluble that need to be mixed together. An example of pharmaceutical applications that use ethanol and water are nasal and oral sprays (Kulkarni & Shaw, 2016). Furthermore, ethanol is widely used as a co-solvent in many oral formulations as a co-solvent and its concentration varies between 0.6 % v/v to 76 % v/v in liquid medications. (Noa Fine-Shamir & Dahan, 2024)

For example, naloxone hydrochloride nasal spray, also known by its brand name KLOXXADO, uses 20% w/w dehydrated alcohol (ethanol) (FDA, 2021). This is a nasal spray used to treat patients suspected of having an opioid overdose. The main API, naloxone HCL, is actually very water soluble and 96% soluble in alcohol but ethanol cosolvent is used as a permeation enhancer which helps the drug penetrate through cell membranes better (Wermeling, 2012).

Another example is nitroglycerin lingual spray, also known as Nitrolingual Pumpspray, which contains 20% v/v ethanol (FDA,2008). Nitrolingual Pumpspray is an oral spray that is sprayed under the tongue to relieve chest pain for patients with angina. The API, nitroglycerin, is

only slightly soluble in water and more soluble in alcohol.(Furman, 2018)Thus the 20% v/v ethanol is a cosolvent that will help nitroglycerin dissolve better.

This report will solely focus on blending a liquid-liquid system and the design parameters are determined based on the defined system. A common co-solvent used in the pharmaceutical industry is water and ethanol, due to their miscibility, compatibility and ability to enhance API solubility. To ensure that the final pharmaceutical product is homogenous and stable cosolvents should be mixed together first. Ensuring that the cosolvents are uniform will ensure that the API and excipients dissolve evenly. (Dohrn et al., 2020) This is important for patient safety because if the API and excipients do not mix uniformly, the drug dosage will be uneven and risks patients taking the wrong dosage.

This report will focus on designing a mixing system of two miscible liquids which will be water and ethanol used in oral formulation and sprays. Based on the ethanol concentration in oral formulations ranging from 0.6-70% v/v (Noa Fine-Shamir & Dahan, 2024) and a commercialized oral spray containing 20% ethanol (FDA, 2021), this report designs this mixing system for a 20% ethanol and 80% water mixture.

2.2 Physical and Chemical Characteristics of Water and Ethanol at 25°C

Table 1: Physical and Chemical Properties of Ethanol and Water

Property	Water (H ₂ O)	Ethanol (C ₂ H ₅ OH)
Polarity	Highly polar (strong dipole)	Moderately polar (OH group + non-polar chain)
Dynamic Viscosity (Pa.s)	0.000890	0.00107
Density (kg/m ³)	997.7	789.8
Molecular Structure	Bent geometry	Linear with hydroxyl group
Miscibility	Fully miscible with ethanol	Fully miscible with water

Water and ethanol are completely miscible in all proportions because they can form strong intermolecular hydrogen bonds with each other. Water and ethanol are both polar molecules and can mix freely with one another in all proportions. Water's hydrogen atoms can bond to ethanol's oxygen atom, and ethanol's hydroxyl group can bond to water molecules. This strong attraction overcomes the slight non-polarity of ethanol's carbon chain. As a result, the two liquids rapidly disperse on a molecular scale once bulk convective mixing brings them into contact. (Sigma-Aldrich, n.d.)

2.3 Equipment Selection

Designing an effective mixing system requires careful selection of vessel geometry, mixer orientation, impeller type, and internal fittings such as baffles. These choices influence flow patterns, mixing time, vortex formation, power input, and mechanical efficiency. The following subsections describe the engineering considerations relevant to stirred tanks used for ethanol-water blending.

2.3.1 Stir Tank vessels

The geometry of the tank and the impeller entrance are crucial aspects to consider while designing a mixing vessel. These considerations are needed for optimal and effective mixing. The most common tank geometry designs are vertical cylindrical tanks and square or rectangular tanks. (Dynamix Inc., n.d.) In the pharmaceutical industry, stainless steel cylindrical tanks are the standard because they provide predictable hydrodynamics, ease of fabrication, and compatibility with Clean-In-Place (CIP) and Sterilize-In-Place (SIP) requirements. Vessel geometry plays a major role in determining circulation patterns, impeller effectiveness, and dead-zone formation. This section provides a brief overview of different possibilities when designing the tank, and the selected specifications will be elaborated on in chapter 3.

2.3.1.1 Geometry of Tank Vessel

i) Cylindrical Tanks

Cylindrical vertical tanks are preferred in pharmaceutical manufacturing for most liquid-phase mixing operations for many reasons such as a uniform flow distribution around the tank, the well understood hydrodynamics in literature, the ease of installing baffles, heating/cooling jackets, and instrumentation, the compatibility with standard impeller configurations, and the efficient CIP and SIP processes.

Typical design ratios of these tanks are in a height-to-diameter ratio (H/D) between 1.0 and 1.5. Impeller clearance is also usually $(C) \approx 0.25\text{--}0.5\text{ D}$. A general cylindrical tank also includes four baffles, each given with a width of $T/10$.

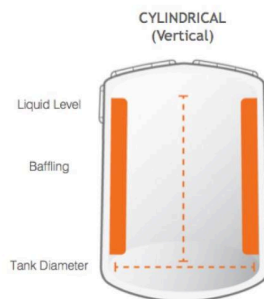


Figure 1: Standard Cylindrical Tank (Dynamix Inc., n.d.)

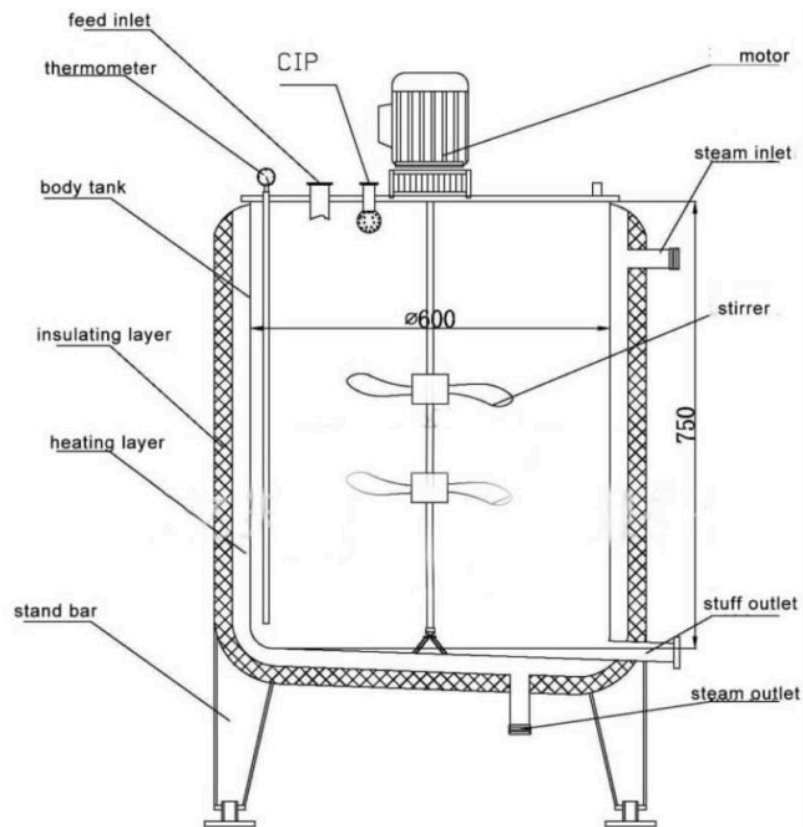


Figure 2: Cylindrical Tank Diagram (Fruit Processing Machine, n.d.)

ii) Rectangular tanks

Rectangular or square tanks are less common in pharmaceuticals due to many issues that are expected with them. These issues include irregular flow patterns, dead zones commonly forming in corners, longer mixing time, and difficult cleaning. These tanks are used mostly for wastewater or low-value blending operations where precision and sterility are not critical.

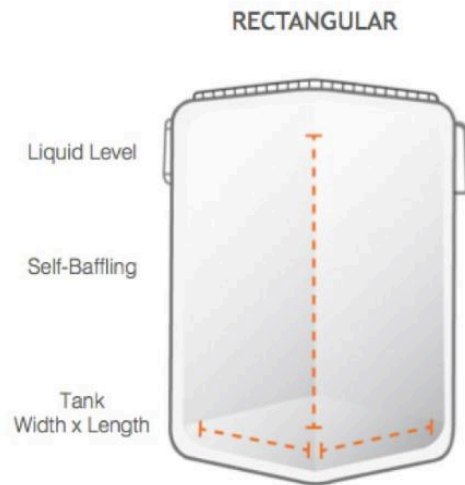


Figure 3: Standard Rectangular Tank (Dynamix Inc., n.d.)

2.3.1.2 Tank Bottom Geometry

There are three main types of tank bottom shapes which include flat, dish (or rounded) and cone. Each has their own advantages and disadvantages based on the mixing systems as outlined in the table below.

Table 2: Advantages and Disadvantages of different Tank Bottom Shapes (MXD Process, n.d.)

Tank Bottom Shape	Advantage	Disadvantage
Flat	- Simple design -Cost effective	-Possible areas of stagnation at the bottom edges of tank -Harder to drain liquid
Dish (rounded)	-Minimized sediment build-up	-Possible product accumulation at the transitional point of the tank

	-Better liquid drainage than flat bottom	- wall and rounded bottom, which can affect proper drainage of liquid if solids are accumulated -Not as cost effective as flat bottom because it is more complicated to manufacture and specialized accessories (drain ports and bottom outlets)
Cone	-Easy to clean -Even better liquid draining than dish bottom	-The most expensive option (complicated to manufacture, needs support structures to stand) -Cone shape possibly cause uneven mixing especially for viscous fluids

i) Flat Bottom Tank

Common applications for flat bottom tanks are industrial processing and storage for water, wastewater, chemicals, fuels, and agricultural liquids such as fertilizers and pesticides. If it is made of stainless steel material it could also be used in pharmaceutical and food processing.



Figure 4: Flat Bottom Cylindrical Tank (MXD Process, n.d.)

ii) Dish Bottom Tank

Dish (rounded) bottom tanks are commonly used in food and beverage manufacturing (particularly for liquids that are likely to have sediment accumulation), chemical and pharmaceutical processing, fermentation tanks in breweries and wineries, storage for wastewater and slurries.



Figure 5: Dish Bottom Cylindrical Tank (MXD Process, n.d.)

iii) Cone Bottom Tank

Cone bottom tanks are frequently used in the manufacturing of chemicals and pharmaceuticals, the storage of agricultural liquids, industrial mixing and blending procedures, fermentation, and brewing.



Figure 6: Cone Bottom Cylindrical Tank (MXD Process, n.d.)

2.3.1.3 Impeller Entrance: Top-Entering vs Bottom-Entering Mixers

In liquid blending applications, mixers can either be installed from the top of the vessel or the bottom. The choice depends on the process requirements, vessel geometry, and desired mixing intensity and hydrodynamics. The orientation of the mixer relative to the tank affects both hydrodynamics and maintenance accessibility.

Top-entering mixers are the most commonly used mixing configuration in liquid blending operations, especially in pharmaceutical, chemical, and food processing industries. In this design, the mixer is mounted on the top head of the vessel, and the impeller shaft extends downward into the tank. This configuration offers excellent flexibility and is widely used for blending miscible liquids such as water and ethanol.

The way that it works is a top-entering mixer is mounted on the top of the tank, with a vertical shaft and impeller submerged in the liquid. When the impeller rotates, it transfers energy to the liquid, creating bulk circulation that moves liquid downward along the shaft and upward along the tank walls. Baffles inside the tank prevent vortex formation and ensure uniform mixing. This mechanism allows efficient blending of low- to medium-viscosity fluids like ethanol and water (Keheng Mixing, n.d.).



Figure 7: Top Entering Mixer

i) Top-Entering Mixers

Top-entering mixers are the most common configuration used in pharmaceutical and industrial liquid blending operations. They are installed on the top of the vessel, with the motor and shaft extending downward into the tank. This arrangement offers several advantages.

Because the drive and seal are located at the top, the risk of contamination or leakage into the batch is reduced, and maintenance is generally easier and more accessible. Top-entry designs are also highly versatile. They can handle a wide range of viscosities, support multiple impellers mounted on the same shaft, and function effectively under turbulent mixing conditions. Their hydrodynamic behavior is well-studied, making performance predictable and reliable (Sealok, 2025).

However, there are also limitations to consider. Top-entering mixers require sufficient vertical clearance above the tank to accommodate the motor, gearbox, and shaft assembly, which may present challenges in facilities with low ceilings or tight equipment spacing. Because the shaft is longer, energy consumption can be slightly higher compared to bottom-entry designs, particularly in large tanks. Additionally, if baffles are not installed, top-entering systems are more prone to vortex formation, which can reduce mixing efficiency and lead to air entrainment (Sealok, 2025).

Top-entering mixers are used in various industries such as pharmaceuticals, food processing, and wastewater treatment. They are commonly used for blending miscible liquids, suspending solids, dissolving powders, maintaining uniformity in storage tanks, and promoting heat transfer in jacketed vessels. Because they accommodate large impellers and work well across a wide range of viscosities, top-entering mixers are the standard choice for general-purpose mixing in medium- to large-scale tanks. They are also preferred when strong axial flow, effective bulk circulation, and flexible impeller configurations are required (Sealok, 2025).

ii) Bottom-Entering Mixers

Bottom-entering mixers are installed at the base of the vessel, with the impeller positioned close to the tank bottom. They are commonly used in industries requiring a high degree of cleanliness and minimal equipment intrusion, such as pharmaceuticals, biotechnology, and food processing.

A bottom-entering mixer is installed at the base of the tank, with the impeller positioned very close to the bottom surface. During operation, the vessel is first filled to an appropriate level to ensure the impeller is fully submerged. When the mixer is activated, the bottom-mounted impeller generates strong upward flow that pushes liquid from the bottom of the tank toward the top, creating an efficient circulation loop. This upward pumping action helps prevent settling, improves contact between ingredients, and ensures uniform blending throughout the vessel. During the mixing process, temperature and pressure can be controlled through external jackets or integrated sensors, depending on the pharmaceutical requirements. Once the desired mixing time and homogeneity are achieved, the product can be discharged directly from the bottom outlet, which is located near the mixer. After production, bottom-entering mixers support effective cleaning and sanitation, as they are fully compatible with Clean-in-Place (CIP) and Sterilize-in-Place (SIP) procedures, allowing the impeller and surrounding surfaces to be thoroughly cleaned without disassembly.

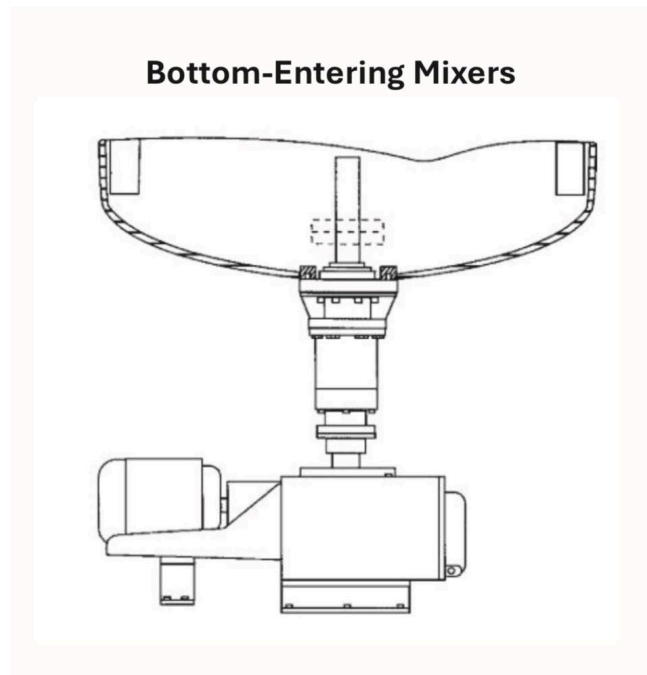


Figure 8: Bottom Entering Mixer

Selecting between a top-entering and bottom-entering mixer depends on the process requirements, tank geometry, sanitation needs, and fluid characteristics. The following factors should be considered:

The table below shows the important factors that must be taken into consideration when selecting a top or bottom entering mixer.

Table 3. Key Factors Influencing the Selection of Mixer Entry Type in Pharma Mixing Systems

Factor	Top-Entering Mixer	Bottom-Entering Mixer
Tank Size and Available Headspace	- Requires headspace above the tank for the motor - Tall, large-diameter tanks	- Ideal for when headspace is limited - Shorter, compact vessels
Compatibility with Multiple Impellers	- Able to support multiple impellers along the shaft	- Limited to a single impeller due to the short shaft length
Energy Efficiency	- Good for large volumes - Requires baffles to avoid vortexing	- Ideal for low-viscosity fluids
Fluid Properties (Viscosity, Density, Volatility)	- Good for a wide viscosity range	- Best for low-viscosity - Less effective for very high-viscosity fluids due to limited shaft length
Risk of Air Entrainment	- Higher risk of vortex formation and air entrainment, especially in unbaffled tanks	- Minimal risk of air entrainment as the impeller is located away from the free surface
Mixing Purpose	- Suitable for blending, solids suspension, homogenization, and large-scale batch mixing. - Versatile and standard in most industrial tanks.	- Ideal for maintaining suspension at the bottom, mixing small batches, or applications requiring minimal disturbance at the liquid surface. - Common in pharmaceutical CIP/WFI tanks.

Maintenance Accessibility	- Easy access for seal and motor service as the equipment is located outside and above the tank	- Difficult to access as the drive system is at the bottom - Requires tanks drainage and entry for major maintenance
Cleanability and Sanitation Requirements	- CIP capable but longer shaft increases cleaning needs	- Very CIP/SIP friendly; minimal internal hardware
Safety and Seal Requirements	- Standard seals; some risk of leakage onto top platform	- Requires high-grade sanitary seals or magnetic drives

2.3.2 Impellers

There are five main types of impellers that are commonly used which include axial flow impellers, radial flow impellers, high shear impellers, close clearance impellers and up/down flow impellers.

Table 4: Summary of Impeller Options

Type	Purpose	Advantages	Disadvantages	Examples	Decision
Axial Flow	Blending, Solid suspension, miscible liquid mixing	Ideal for low-viscosity miscible liquids	Not suitable for high-viscosity liquids	-Propeller -Pitched blade turbine -Hydrofoil	Accepted
Radial Flow	Low to	Ideal for	Can cause	-Flat-blade	Rejected

	medium viscous fluids, gas dispersion	gas-liquid and liquid-liquid dispersion	stratification or compartmentalization in the mixing tank, high power consumption	impeller - Rushton turbine - Hollow-blade turbine	
High Shear	Emulsification, Grinding, pigment dispersion	Intense shear	Not suitable for simple blending, can cause heating	-Flat-blade Impeller -Disk Turbine (Rushon) -Hollow-Blade Turbine (Smith)	Rejected
Close-clearance	Mixing of highly viscous fluids	-Meant for high viscous fluids	-More expensive design	-Anchor Impeller -Helical Ribbon Impeller	Rejected
Up/Dpwn Flow	Mixing where direction control matters	Can control flow direction	Low efficiency compared to axial designs	-Disk -Plate -Circular	Rejected

The impeller chosen for this project is the axial flow impeller because they are best for mixing based on needing a high bulk flow. Miscible liquids already naturally will want to dissolve with each other therefore high shear impellers like radial flow impellers will not be needed.

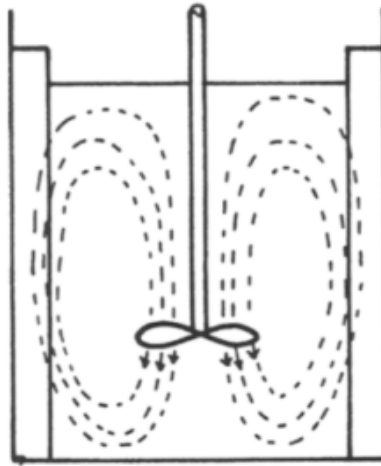


Figure 9: Axial Flow Impeller

Axial flow impellers are widely used in liquid-liquid blending because they generate a strong top-to-bottom pumping action that promotes rapid bulk circulation. This flow pattern is ideal for miscible liquid blending, such as water-ethanol mixing, where the objective is uniform distribution rather than high shear. The following are the main types of axial flow impellers used in industrial mixing systems.

2.3.2.1 Marine Propeller

Marine propellers are high-efficiency axial flow impellers consisting of two or three curved blades similar to a boat propeller. They produce a strong downward pumping action and are commonly used in low-viscosity fluids.

Marine propellers are high-efficiency axial-flow impellers designed to generate strong pumping action with minimal power consumption. They are particularly effective in

low-viscosity liquids such as water and ethanol, where they provide high axial flow and relatively low shear compared to turbine-style impellers. Their compact design also allows them to operate at higher rotational speeds, making them suitable for applications requiring rapid circulation.

These impellers offer several advantages. They provide excellent performance for fast blending and homogenization, are simple in design, cost-effective to manufacture, and deliver good overall energy efficiency. However, marine propellers also have disadvantages. They are not suitable for medium- or high-viscosity fluids, and they have limited capability when handling solids or gas-dispersion processes, where higher shear or more specialized flow patterns are required. Marine propellers are often used in small to medium-sized tanks or applications requiring fast circulation but not high shear.



Figure 10: Marine Impeller

2.3.2.2 Pitched Blade Turbine

Pitched-blade turbines are versatile and widely used for liquid mixing. They also create an axial discharge. Most pitched-blade turbines have four blades, mounted at a 45° angle, and with an actual blade width one-fifth of the impeller diameter. For hydraulic efficiency of a four blade impeller with 45° pitch angles, the optimal arrangement of baffled mixing system is recommended to have D/T (ratio of diameter of impeller to diameter of tank) and C/T (ratio of clearance to tank diameter) are equal to $\frac{1}{3}$. (Fořt, 2010)

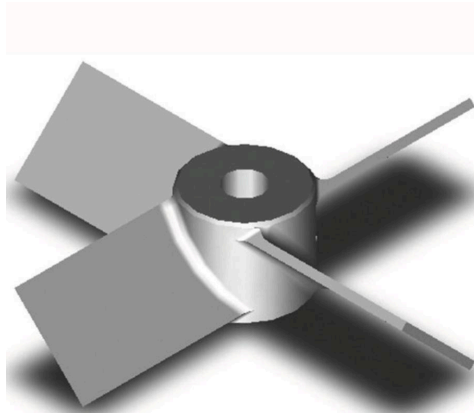


Figure 11: Pitched-Blade Turbine

Pitched-blade turbines are widely used in liquid-liquid mixing because of their versatility and balanced hydrodynamic performance. With blades typically pitched at 45° , these impellers generate a mixed flow pattern consisting of both axial and radial components, making them suitable for a broad range of blending applications. The blades usually have a width equal to approximately 20% of the impeller diameter, which helps provide moderate shear levels while maintaining efficient pumping.

These impellers offer several advantages. They create strong top-to-bottom circulation, which is essential for achieving uniform mixing in miscible liquid systems such as ethanol-water solutions. Pitched-blade turbines also perform well across low to medium viscosity fluids and are capable of suspending light solids when necessary. Additionally, they can be configured for either upward or downward pumping depending on process requirements.

However, pitched-blade turbines also have some limitations. They tend to consume more power than marine propellers and generate slightly higher shear levels, which may be unnecessary for simple miscible liquid blending where high shear is not required. Despite these considerations, their overall balance of pumping capacity, flow control, and versatility makes them a common choice in industrial mixing applications.

For this design, a pitched-blade turbine is particularly appropriate because it produces sufficient axial pumping to blend ethanol-water while maintaining gentle shear that prevents excessive energy use. To be effective, axial flow impellers should be designed to pump downward toward the bottom of the vessel. Upward pumping impellers do a poor job of lifting flow from the bottom and turning flow at the free surface. However, in some situations, such as gas dispersion and floating solids mixing, up-pumping may be more effective. The flow discharge from a pitched blade impeller has components of both axial and radial flow velocity in low to medium viscosity liquids, and is considered to be a mixed-flow impeller.

2.3.2.3 Hydrofoil Impellers

Hydrofoil impellers (e.g., Lightnin A310/A510 designs) are high-efficiency axial-flow impellers engineered to maximize pumping while minimizing power draw. The blades have a streamlined airfoil profile that reduces drag and shear.

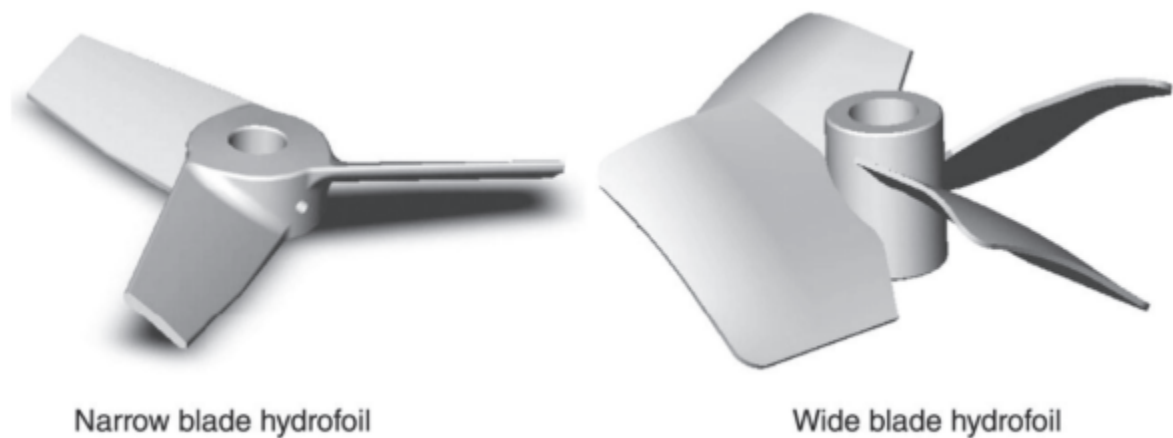


Figure 12: Hydrofoil Impeller

Hydrofoil impellers are engineered with thin, curved blades resembling an aircraft wing, allowing them to deliver exceptionally high flow rates while consuming significantly less power than other impeller types. Their aerodynamic shape enables them to generate strong bulk motion with minimal turbulence, resulting in smooth and efficient circulation throughout the tank.

These impellers offer several advantages. They are highly energy-efficient, often described as providing “high flow, low shear,” making them ideal for large-scale blending operations where power savings are important. Their smooth flow patterns also help reduce

vortex formation and are especially beneficial when handling delicate or temperature-sensitive materials that could be damaged by high shear or excessive turbulence.

Despite their strengths, hydrofoils have limitations. They are not suitable for very high-viscosity fluids, where more robust or higher-shear impellers are required. Additionally, hydrofoils tend to be more expensive than marine propellers or pitched-blade turbines due to their more complex blade geometry and manufacturing requirements. Hydrofoils are commonly used in large pharmaceutical mixing vessels where precise control, low shear, and energy efficiency are required.

Axial flow impellers are advantageous in miscible liquid blending operations because they promote rapid circulation and uniform composition with minimal shear. Marine propellers provide fast blending at low cost, pitched-blade turbines offer versatility and strong mixed flow, and hydrofoils maximize efficiency with gentle turbulence. For a water-ethanol mixing system, the pitched-blade turbine provides the optimal balance of pumping capacity, mixing time, and energy efficiency.

2.3.3 Baffles

Baffles are vertical plates installed inside the tanks, piping systems or vessels to control fluid flow, prevent dead zones, increase turbulence and improve mixing efficiency. In pharmaceutical liquid-liquid blending operations, baffles are essential for achieving uniform circulation patterns, enhancing turbulence, and ensuring batch consistency. In unbaffled tanks, liquids tend to swirl around the impeller and form a vortex, which reduces the mixing

performance and introduces air into the product. This can cause problems in pharmaceutical processes where aeration can cause foaming, oxidation or inconsistency.

Baffles play a critical role in ensuring efficient and consistent mixing performance in pharmaceutical stirred tanks. One of their primary functions is to direct fluid flow. Without baffles, liquid tends to spin in a circular pattern around the impeller, resulting in poor mixing and limited contact between fluid layers. By interrupting this swirling motion, baffles redirect the liquid into more effective axial (top-to-bottom) and radial (outward) flow paths. This structured flow promotes uniform distribution of ingredients, eliminates dead zones, and ensures consistent blending of APIs, solvents, and excipients (Connor, 2024).

Baffles also enhance turbulence within the tank. By breaking up rotational flow, they increase local fluid velocity fluctuations, which improves mass transfer and accelerates blending. This is especially important in pharmaceutical operations that require molecular-level homogeneity, such as solvent preparation or uniform API distribution. In addition, baffles shape the overall hydrodynamic environment of the tank by reinforcing both axial and radial flow components. Axial flow supports vertical circulation, while radial flow distributes liquid outward toward the tank walls. Together, these motions ensure complete circulation throughout the vessel and prevent stagnant regions, which is essential for miscible liquid blending.

Another important function of baffles is preventing vortex formation. A vortex can draw air into the liquid, causing issues such as oxidation, foaming, inconsistent density, and potential contamination which can be unacceptable in some pharmaceutical processing. By suppressing vortexing, baffles help the impeller generate efficient, entrainment-free flow.

Beyond their mixing benefits, baffles also support sanitary and hygienic design. In pharmaceutical manufacturing, they are typically constructed from polished stainless steel with smooth surfaces and sanitary welds. This prevents material buildup, ensures compatibility with CIP and SIP systems, and helps maintain compliance with GMP guidelines. Finally, baffles improve batch-to-batch consistency by stabilizing the flow pattern and ensuring that each batch experiences similar mixing conditions. By reducing vortexing, eliminating dead zones, and promoting uniform energy distribution, they help ensure repeatable product quality and reliable performance across multiple production cycles.

The number and arrangement of baffles play an important role in achieving efficient and stable mixing performance. In most cylindrical pharmaceutical mixing tanks, four vertical baffles are considered the industry standard because they provide the best balance between flow control, turbulence generation, and ease of cleaning. Each baffle is typically installed equally spaced around the tank wall and sized at about 1/10 of the tank diameter to effectively interrupt swirling without causing excessive resistance or power consumption. Fewer than four baffles can lead to incomplete suppression of vortexing and uneven flow patterns, while adding more than four offers little improvement and may complicate tank cleaning or increase energy use. Proper configuration ensures that the impeller's energy is converted into strong axial and radial flow, improving mixing uniformity and reducing dead zones.

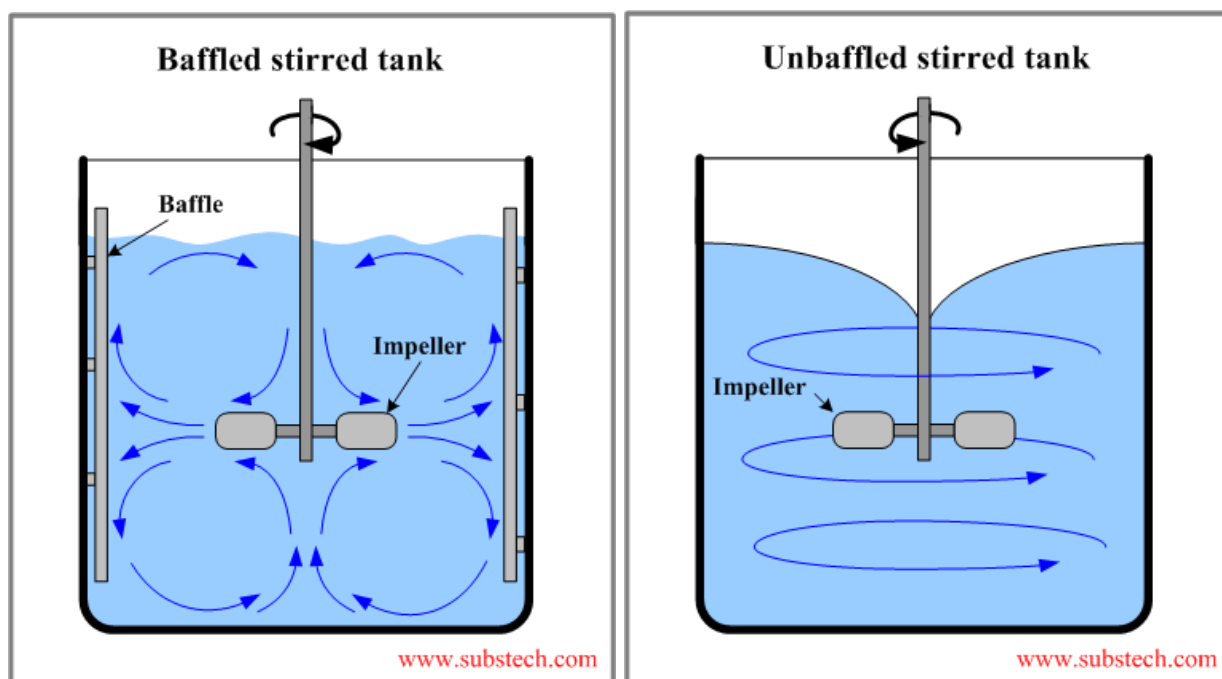


Figure 13: Baffled vs. Unbaffled Tank

Baffles have a significant impact on overall mixing performance because they directly shape the flow behavior inside the tank. By disrupting the circular motion that naturally forms around a rotating impeller, baffles convert rotational energy into effective top-to-bottom and side-to-side fluid movement. This results in faster blending, shorter mixing times, and more uniform concentration throughout the tank. Baffles also increase turbulence, which enhances mass transfer and helps eliminate stagnant or slow-moving areas where ingredients may otherwise accumulate. In addition, the use of baffles reduces vortex formation and air entrainment, preventing issues such as foaming, oxidation, and inconsistent product quality, which are critical in pharmaceutical processes. Overall, the presence of baffles makes the mixing process more efficient, predictable, and controlled, which supports consistent batch quality and reliable system performance.

Baffles significantly influence the power consumption by the impeller during liquid blending. In an unbaffled tank, most of the energy is lost to swirling and formation of a central vortex. Mixing is not as effective because the fluid is rotating with the impeller.

When baffles are installed, they interfere with the rotational motion, converting it to more useful turbulent flow, especially radial and axial mixing. This increases the resistance the fluid presents to the impeller blades, which raises power consumption and results in a higher power number, (N_p). However, the additional energy is used more efficiently, since it is directed toward creating turbulence rather than sustaining a vortex. As a result, baffled systems achieve better mixing performance and provide more predictable behaviour, allowing engineers to apply well-established power correlations for accurate scale-up and design.

For typical fully baffled tanks ($4 \times$ baffles at $\sim 1/10$ tank diameter), the power number becomes effectively constant with Reynolds number in the turbulent regime, making power consumption predictable and controllable.

Chapter 3. Description of Chosen Design

3.1 Overall Design Process

The design process began by clearly defining the mixing system and its intended application. This system is used for liquid blending of miscible liquids, specifically the preparation of a water-ethanol mixture. This pre-blending step is essential before adding APIs and excipients, as ethanol must be diluted to an appropriate concentration to ensure proper

dissolution. A 20% v/v ethanol-water mixture was selected based on the ethanol content found in a commercial oral angina spray, Nitrolingual Pumpspray (FDA, 2008).

Following this, the relevant fluid properties were identified to guide the design. Key characteristics such as density, viscosity, miscibility, polarity, and molecular structure for both ethanol and water were evaluated. The primary goal of the mixing system was to achieve rapid blending and produce a uniform, almost completely homogeneous solution. To support this objective, the system was designed to operate under turbulent flow conditions to promote strong circulation and shorten mixing time.

Once the process goals and fluid characteristics were established, appropriate equipment was selected. A cylindrical, flat-bottom tank was chosen due to its predictable hydrodynamics and suitability for pharmaceutical applications. The agitation system consists of a four-blade pitched-blade turbine with a 45° blade angle, selected for its ability to generate strong axial flow ideal for miscible liquid blending. To prevent vortexing and improve mixing efficiency, the tank was fitted with four vertical baffles, each with a width equal to one-tenth of the tank diameter.

Sizing calculations were then carried out based on an assumed industrial-scale working volume of 2000 L (2 m^3), a typical capacity for medium-scale pharmaceutical production (PPMS, 2025). Key design parameters included the dimensions of the tank, impeller, and baffles; the impeller-to-tank diameter ratio; the required impeller speed; and the power input necessary to achieve the desired mixing intensity. Additional engineering parameters including Reynolds number, power number, pumping capacity, Froude number, and predicted mixing time were

evaluated to confirm that the chosen configuration provides efficient, reliable, and rapid mixing consistent with the system requirements.

3.2 Summary of Selected Equipment

3.2.1 Chosen Mixing Vessel

The selected mixing vessel for this system is a cylindrical, flat-bottom tank. This geometry is well suited for pharmaceutical liquid blending because it provides predictable hydrodynamics, minimizes dead zones, and supports strong axial circulation when used with an appropriate impeller. Compared to rectangular tanks, cylindrical vessels experience fewer stagnant regions and offer more uniform flow patterns, which is important for achieving rapid homogeneity in miscible liquid systems.

A flat-bottom design was chosen for its simplicity, cost-effectiveness, and ease of fabrication. For the ethanol-water mixture considered in this project, the absence of suspended solids or sediment eliminates concerns about material buildup at the bottom edges of the tank. Since both liquids are low-viscosity and fully miscible, they blend readily without requiring specialized bottom geometry. Additionally, flat-bottom cylindrical tanks are easier to equip with baffles, clean, sanitize, and integrate with CIP/SIP procedures making them fully compatible with pharmaceutical manufacturing requirements.

Overall, the cylindrical flat-bottom tank offers the most practical, economical, and operationally efficient vessel configuration for this mixing system.

3.2.2 Chosen Impeller

The impeller selected for this mixing system is a four-blade pitched-blade turbine with a 45° blade angle, designed to generate strong axial flow. After evaluating multiple impeller categories, axial-flow designs were identified as the most suitable for this application. For low- to medium-viscosity fluids such as ethanol and water, the primary comparison is between radial-flow and axial-flow impellers. Radial impellers create high shear but relatively lower bulk circulation, whereas axial impellers produce strong top-to-bottom pumping with lower shear. Since ethanol and water are fully miscible and readily dissolve in one another, high shear is not required. Instead, the priority is rapid circulation to reduce mixing time and ensure fast, uniform blending. This makes axial-flow impellers the optimal choice.

Among axial-flow options, three designs were compared which were marine propellers, pitched-blade turbines, and hydrofoils. Marine propellers are inexpensive and provide fast blending but offer limited versatility. Hydrofoils are highly energy-efficient and produce smooth, low-shear flow but are more costly and typically used in large-scale or shear-sensitive processes. The pitched-blade turbine provides the best balance of performance characteristics as it delivers strong axial pumping, moderate shear, excellent circulation, and reliable mixing performance across various operating conditions. For these reasons, the 45° pitched-blade turbine with four blades was selected as the impeller for this system. It offers an effective combination of pumping capacity, energy efficiency, and mixing speed, making it the most appropriate choice for producing a homogeneous ethanol-water mixture.

3.2.3 Chosen Baffles

Four vertical baffles were selected for this design because they provide the most effective and efficient mixing performance in a standard cylindrical tank. A four-baffle configuration interrupts the swirling motion generated by the impeller, preventing vortex formation and promoting stronger axial and radial flow throughout the vessel. This results in faster mixing, improved circulation, and greater uniformity in concentration and temperature.

Using fewer than four baffles often leads to inadequate flow control, increased swirling, and the formation of dead zones. Conversely, adding more than four baffles typically does not enhance performance and can unnecessarily increase energy consumption and cleaning complexity. For most liquid-blending operations, particularly for low-viscosity miscible systems such as ethanol-water, four baffles are considered the industry standard. This configuration ensures reliable, predictable, and consistent mixing behavior, making it the most appropriate choice for the system designed in this report.

Chapter 4. Mixing System Design and Characterization

4.1 Design Basis and Operating Conditions

The mixing system is designed to prepare a 20% v/v ethanol-water solution under batch, isothermal operating conditions. A total batch volume of 2.0 m³ is processed in a vertical, baffled cylindrical tank equipped with a mechanically-driven impeller. The objective of the mixing operation is to achieve homogeneity that is close to completeness within 25-30s of mixing,

consistent with typical requirements for miscible liquid blending. The mixing vessel and all wetted parts (tank shell, baffles, impeller, and nozzles) are made with 316L stainless steel. This material offers excellent corrosion resistance in the ethanol-water system, is compatible with typical cleaning and sterilization procedures, and is widely used in sanitary mixing applications, ensuring safety, good cleanability and mechanical robustness (CHENMA Machinery, n.d.).

To ensure rapid and reliable mixing, a design power input of 0.75 kW/m^3 was selected, which lies within the recommended range for vigorous liquid-liquid blending of low-viscosity systems (Oldshue, 1983). The operation is assumed to be isothermal, supported by the low predicted temperature rise due to agitation (which will be further expanded on in section 4.5).

The system is operated with a 300 s batch time, which includes filling, mixing, and discharge. Agitation is provided by a 45° 4 blade pitched-blade turbine (PBT) operating at a nominal speed to be determined through hydrodynamic calculations (Sections 4.3 and 4.5). The mixing tank is fully baffled to promote turbulent flow and eliminate vortex formation.

Multiple assumptions are used in this design such as the total working volume being 2.0 m^3 . A target composition of 20% v/v ethanol and 80% v/v water was chosen because that is the standard nasal spray composition. The operating temperature is at 22°C and is isothermal. The required mixing quality is almost complete homogeneity (approximately 99%) with a desired agitation intensity of 0.75 kW/m^3 . It is also assumed that it is a fully baffled tank, four baffles each of width $T/10$.

4.2 Fluid Properties and Mixture Composition

The working fluid in this process is a 20% v/v ethanol-water mixture prepared at 22 °C.

Physical properties of the pure components were obtained from standard thermophysical databases, and mixture properties were calculated assuming ideal mixing due to the low viscosity and complete miscibility of the components. It is important to note that there is a flammability risk which is negated by ensuring proper grounding, and the use of explosion proof motors that are ATEX-rated.

At 22 °C, water and ethanol have densities of 997.7 kg/m³ and 787.8 kg/m³ (Engineering Toolbox, 2001), respectively. The dynamic viscosity of the mixture was taken as 1.5×10^{-3} Pa·s (Engineering Toolbox, 2001), which is representative of low-alcohol aqueous solutions under ambient conditions. The specific heat capacity of a 20% ethanol mixture is approximately 3.7 kJ/(kg·K) (Engineering Toolbox, 2001).

Based on a total batch volume of 2.0 m³, the volumetric and mass contributions of each component were calculated. The mixture contains 0.40 m³ of ethanol and 1.6 m³ of water, corresponding to masses of 315.12 kg and 1596.32 kg, respectively. This results in a total mixture mass of 1911.44 kg with ethanol and water mass fractions of 0.165 and 0.835. The resulting mixture density is 955.72 kg/m³ (Engineering Toolbox, 2001) at 22°C.

The low viscosity and moderate density of the mixture ensure operation in the fully turbulent regime for the selected impeller speed (expanded on in Section 4.4). These properties directly influence the Reynolds number, power consumption, and expected mixing time.

4.3 Mixer Geometry and Configuration

The mixing system consists of a vertical, baffled, flat-bottom cylindrical tank equipped with a 45° 4 blade pitched-blade turbine. The tank geometry and impeller configuration were selected based on standard mixing design guidelines to ensure efficient axial-radial flow, strong top-to-bottom circulation, and operation in the turbulent regime.

The tank has an internal diameter of 1.40 m and a working liquid height equal to the diameter ($H/T = 1.0$), which is commonly used for general-purpose liquid blending. The total working volume is 2.0 m^3 . Four baffles are installed on the tank walls, each with a standard width of $T/10 = 0.14 \text{ m}$, to suppress vortex formation and maintain fully turbulent flow patterns.

Agitation is provided by a 45° pitched-blade turbine with four blades, selected for its strong axial pumping and suitability for miscible liquid blending. The impeller diameter is $D = 0.467 \text{ m}$, giving a tank diameter to impeller diameter (D/T) ratio of $1/3$, which lies within the recommended 0.3 - 0.5 range for effective blending. The width of the impeller is found using the common impeller width to diameter ratio of $W/D = 1/5$. Solving for the impeller width gives 0.093 m . The impeller clearance was set using the ratio of $C/T = 1/3$, which lies at the upper end of the typical design guideline range (0.20 - 0.30). Although slightly higher than the accepted range of 0.2 - 0.3 , this clearance remains acceptable for low-viscosity miscible liquid blending and provides adequate axial circulation without risk of vortex formation. The impeller is positioned at a clearance of 0.467 m from the tank bottom, providing adequate bottom sweeping and preventing dead zones.

The tanks are usually never filled to the very top, so assuming an empty space of around 0.4m, the total tank height would be 1.8m. The impeller shaft diameter was selected as 0.0373 m (using the ratio of shaft diameter to impeller diameter of 2/25), which lies within typical design practice for mechanically driven mixers and provides sufficient mechanical rigidity and torque capacity for the calculated power requirement (found in section 4.5).

It is important to note that all the ratios that were selected allow the usage of a constant power number which will be discussed in chapter 4.4. The table below outlines the chosen geometric ratios and outlines the acceptable ranges for this operation.

Table 5: Calculated ratios and their acceptable ranges

Ratio	Calculated Value	Acceptable Range
D/T	1/3	0.3-0.5
C/T	1/3	0.2-0.3
H/T	1	1
d/D	2/25	0.03-0.08
W/D	1/5	0.15-0.25

The ratios presented in the table above ensure strong axial circulation, efficient convective blending, and predictable hydrodynamic behavior for the subsequent power and mixing-time analysis.

This configuration promotes high pumping capacity, uniform top-to-bottom flow, and rapid homogenization of low-viscosity liquids. The selected geometry directly influences the

power number, pumping number, and ultimately the mixing time, all of which are evaluated in Section 4.4 and Section 4.5. The table and figure below show all the dimensions of the tank.

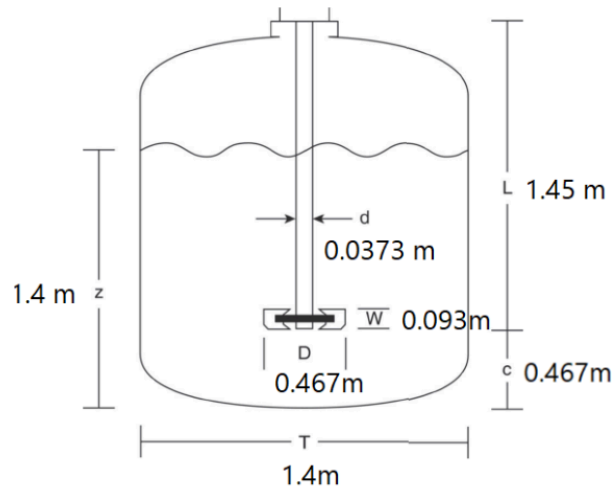


Figure 14: Geometry of Tank

Table 6: Tank Dimensions

Parameter	Value
Tank Diameter	1.4 m
Liquid Height	1.4 m
Total Tank Height	1.8 m
Impeller Diameter	0.467 m
Impeller Width	0.0933 m
Baffle Width	0.14 m
Shaft Diameter	0.0373 m
Clearance	0.467 m

4.4 Hydrodynamic Regime (Dimensionless Numbers)

The hydrodynamic behavior of the mixing system was evaluated using standard dimensionless groups. These parameters characterize the flow regime, energy dissipation, and circulation generated by the impeller, and they directly determine the power consumption and mixing performance.

In order to calculate Reynolds number, first the impeller speed must be determined. To do so, the design power input of 0.75 kW/m^3 is used. This is a ratio of Power/Volume, so multiplying it will result with Power. Using the assumption that it is turbulent (which will be later fact checked after finding Reynolds number), the power number 1.27 is used. Using trial and error, an impeller speed of 3.83 s^{-1} was determined.

The Reynolds number for stirred tanks is defined as $Re = \rho ND^2/\mu$, and using the mixture density of 955.7 kg/m^3 , viscosity of $1.5 \times 10^{-3} \text{ Pa}\cdot\text{s}$, impeller diameter of 0.50 m , and operating speed of 3.83 s^{-1} , the calculated Reynolds number is 5.3×10^5 , and since $Re \gg 10^4$, the system operates in the fully turbulent regime, allowing the use of constant power and pumping numbers for the selected impeller. This also confirms the assumption made earlier.

Because it is a fully turbulent regime, the power number for a 45° pitched-blade turbine with 4 blades is approximately constant. Based on standard agitation design charts, the value for the power number is 1.27. The Froude Number (Fr) reflects the relative importance of inertial and gravitational forces, and is calculated as $Fr = (N^2 D)/g$. Solving gives a Froude Number of 0.69.

This moderate value confirms strong axial pumping without excessive vortex formation (further minimized by baffles).

The pumping number represents the non-dimensional circulation capacity of the impeller. For a 45° pitched-blade turbine in turbulent flow, typical values lie in the range 0.6-0.8. Due to turbulent conditions, 0.79 can be taken as the pumping number.

The table below summarizes all the dimensionless numbers and the information that each value represents.

Table 7: Dimensionless numbers and their Interpretations

Parameter	Value	Interpretation
Reynolds Number (Re)	5.3×10^5	Fully turbulent
Power Number (N_p)	1.27	Constant (due to turbulent flow)
Froude Number (Fr)	0.69	Strong axial flow, no vortex
Pumping Number (N_Q)	0.79	High circulation

These dimensionless numbers confirm that the mixer operates in a fully turbulent, high-circulation regime, enabling the use of standard correlations for power draw and mixing time. The high Reynolds number and strong pumping guarantee rapid homogenization, validating the assumptions made in the design basis.

4.5 Power, Mixing Performance, and Energy

The impeller power requirement, circulation capacity, and mixing time were evaluated using standard correlations for baffled, turbulent mixing systems equipped with pitched-blade turbines. These performance metrics determine whether the selected mixer configuration delivers the desired homogeneity within operational constraints.

The impeller power draw was estimated using The formula of $P = N_p \cdot \rho \cdot N^3 \cdot D^5$ With $N_p = 1.27$, $\rho = 955.7 \text{ kg/m}^3$, $N = 3.83 \text{ s}^{-1}$, and $D = 0.467 \text{ m}$. The calculated power comes out to be 1.5 kW. Dividing this value by the total liquid volume gives a value which is very close to the design power input of 0.75 kW/m^3 , which is in the range for vigorous miscible-liquid blending and is consistent with the design basis ($0.3\text{-}0.8 \text{ kW/m}^3$).

For the pumping capacity, the bulk circulation generated by the impeller was calculated using the pumping number, which was determined to be 0.79. Calculating $Q = N_Q \cdot N \cdot D^3$ gives a value of $0.308 \text{ m}^3/\text{s}$. This ensures rapid convective mixing throughout the vessel.

Mixing time was estimated using the empirical correlation for turbulent baffled tanks using the formula of $t_{99} = k \cdot T^2 / (N \cdot D^3)$ (Paul et al., 2004). With $k = 5$ (which is common for turbulent conditions when D/T is in a range of $0.3\text{-}0.5$, and with 4 baffles with a ratio of $T/10$ as used in this design) (Paul et al., 2004), $T = 1.40 \text{ m}$, $D = 0.50 \text{ m}$, and $N = 3.83 \text{ s}^{-1}$. The total mixing time comes out to be 25.18s. This value represents the time required to reach approximately 99% homogeneity, consistent with standard definitions of mixing time.

The energy required for the mixing portion of each batch is equivalent to the power multiplied by the mixing time. Multiplying the two values gives 38 kJ. The assumption of isothermal conditions can also be made because calculating the change in temperature comes out to be 0.00537K. This value of change in temperature is so small that isothermal condition is a reasonable assumption to make.

Chapter 5. Conclusion

This project developed and analyzed a mixing system for the preparation of a 20% v/v ethanol-80% v/v water co-solvent mixture intended for pharmaceutical applications such as oral and nasal sprays. The primary objective was to design a mechanically agitated tank capable of achieving near-complete homogeneity within a short mixing time, while satisfying sanitary, hydrodynamic, and energy constraints typical of pharmaceutical manufacturing. To meet this objective, a comprehensive review of mixing theory, ethanol-water properties, and common industrial equipment was performed and used to guide the selection of vessel geometry, impeller type, and internal fittings.

Based on this review, a vertical, flat-bottom cylindrical tank constructed from 316L stainless steel was selected due to its well-understood hydrodynamics, compatibility with CIP and SIP procedures, and widespread use in pharmaceutical blending. A fully baffled configuration with four baffles (width $T/10$) was chosen to prevent vortex formation, enhance turbulence, and ensure uniform axial and radial circulation. For agitation, a 45° four-blade pitched-blade turbine (PBT) was selected as the impeller, since it provides strong axial pumping,

efficient bulk circulation, and sufficient mixing intensity for low-viscosity miscible liquids without unnecessary shear.

The chosen geometry for a 2.0 m³ working volume ($T = 1.40$ m, $H/T = 1.0$, $D/T = 1/3$, $C/T = 1/3$, $d/D = 2/25$) lies within accepted design guidelines for general-purpose liquid blending. Using ethanol-water mixture properties at 22 °C, the hydrodynamic regime was characterized by a Reynolds number on the order of 5.3×10^5 , confirming fully turbulent operation. This allowed the use of constant power and pumping numbers for the PBT ($N_p = 1.27$, $N_Q = 0.79$). At an impeller speed of 3.83 s^{-1} , the design achieves a power input of approximately 1.5 kW, corresponding to 0.75 kW/m^3 , which falls within the recommended range for vigorous liquid-liquid blending of low-viscosity systems. The predicted pumping capacity of $0.308 \text{ m}^3/\text{s}$ ensures strong bulk circulation throughout the vessel.

Mixing performance calculations indicate that the system can achieve approximately 99% homogeneity in about 25 s. The associated energy input per batch is about 38 kJ, and the corresponding temperature rise in the fluid is negligible. These results demonstrate that the selected vessel-impeller-baffle combination can deliver fast, efficient, and reproducible blending suitable for pharmaceutical co-solvent preparation, while also remaining energy-conscious and compatible with hygienic design requirements.

Although the design meets the project objectives at the conceptual and calculation level, several aspects could be refined in future work. Experimental validation using tracer studies or conductivity measurements would be valuable to confirm the predicted mixing time and assess any scale-dependent effects. Computational fluid dynamics could be used to visualize flow

patterns, identify potential dead zones, and optimize baffle or impeller placement. In addition, evaluating the sensitivity of mixing performance to changes in ethanol concentration, viscosity, or batch volume would support robust scale-up and flexible operation for different formulations. Finally, integration with process control strategies (e.g., variable-speed drives or feedback-based mixing endpoints) could further improve consistency and efficiency in an industrial setting.

Overall, this project demonstrates that a properly designed stirred tank equipped with a pitched-blade turbine and standard baffling can provide rapid, energy-efficient, and hygienic blending of ethanol-water co-solvents for pharmaceutical applications. The methodology used-linking fluid properties, dimensionless analysis, and engineering design criteria-can be extended to other miscible liquid systems and serves as a useful framework for future mixing system designs in the pharmaceutical industry.

Nomenclature

API - Active Pharmaceutical Ingredient

PBT - Pitched-blade Turbine

CIP - Clean-In-Place

SIP - Sterilize-In-Place

API – Active Pharmaceutical Ingredient

CIP – Clean-In-Place

C – Clearance from bottom (impeller-to-tank diameter ratio,

d – Impeller shaft diameter [m]

D – Impeller diameter [m]

D/T – Tank diameter to impeller diameter ratio

Fr – Froude number (dimensionless)

GMP – Good Manufacturing Practice

H/T – Liquid height to tank diameter ratio

k – Mixing time constant, dimensionless

m – Mass [kg]

N – Impeller rotational speed [1/s]

N_p – Power number (dimensionless)

N_Q – Pumping number (dimensionless)

P – Impeller power input [W or kW]

PBT – Pitched-Blade Turbine

Q – pumping capacity [m^3/s]

Re – Reynolds number (dimensionless)

RPM – Revolutions per minute

SIP – Sterilize-In-Place

T – Tank diameter [m]

t – Mixing time [s]

V – Volume [m^3]

W – Width [m]

ρ – Density [kg/m^3]

μ – Dynamic viscosity [$\text{Pa}\cdot\text{s}$]

ν – Kinematic viscosity [m^2/s]

C_p – Specific heat capacity [$\text{kJ}/\text{kg}\cdot\text{K}$]

ΔT – Temperature change [K]

η – Mixing efficiency (dimensionless)

W/W – Weight/weight concentration

V/V – Volume/volume concentration

API – Active Pharmaceutical Ingredient

PBT – Pitched-Blade Turbine

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Appendix A.

Power Numbers of Various Impellers under Turbulent Conditions with Four Standard Baffles

Impeller Type	N_p
Concave- or hollow-blade turbine	4.1
Ekato MIG—3 impellers, $D/T = 0.7$	0.55
Ekato Intermig—2 impellers, $D/T = 0.7$	0.61
High-shear disk at $Re = 10\,000$	0.2
	(lower for lower Re)
Lightnin A310	0.3
Chemineer HE3	0.3
The following are all for $D = T/3$, $C = T/3$, and blade width $W = D/5$:	
45°PBT; 4 blades	1.27
45°PBT; 6 blades	1.64
Marine propeller (1.0 pitch)	0.34
Marine propeller (1.5 pitch)	0.62
Smith or concave- or hollow-blade with 6 blades	4.4

Pumping Number, N_Q , under Turbulent Conditions for Various Impellers

Impeller Type	N_Q
Propeller	0.4–0.6
Pitched blade turbine	0.79
Hydrofoil impellers	0.55–0.73
Retreat curve blade	0.3
Flat-blade turbine	0.7
Disk flat-blade turbine (Rushton)	0.72
Hollow-blade turbine (Smith)	0.76

Appendix B. Sample calculations

Fluid Properties and Mixture Composition Calculations

The calculations below are shown for ethanol, but the same method applies to water using its corresponding properties.

Component Volumes

$$V_{ethanol} = \%v/v_{ethanol} * v_{total} = 0.2 * 2m^3 = 0.4m^3$$

Mass of Components and Overall Mass

$$m_{ethanol} = \rho_{ethanol} * V_{ethanol} = 787.8 \frac{kg}{m^3} * 0.4m^3 = 315.12kg$$

$$m_{total} = m_{water} + m_{ethanol} = 1596.32kg + 315.12kg = 1911.44kg$$

Mass Fraction of Components

$$w_{ethanol} = \frac{m_{ethanol}}{m_{total}} = \frac{315.12kg}{1911.44kg} = 0.165$$

Overall Density

$$Mixture\ Density = \frac{m_{total}}{V} = \frac{1911.44kg}{2m^3} = 955.72 \frac{kg}{m^3}$$

Mixer Geometry and Configuration Calculations

Tank Geometric Ratios

$$\frac{D}{T} = \frac{0.467m}{1.4m} = 0.33$$

$$Clearance = \frac{T}{3} = \frac{1.4m}{3} = 0.467m$$

$$\frac{C}{T} = \frac{0.467m}{1.4m} = 0.25$$

$$\frac{H}{T} = \frac{1.4m}{1.4m} = 1$$

$$Baffle Width = \frac{T}{10} = \frac{1.4m}{10} = 0.14m$$

$$Impeller Width = \frac{W}{D} * D = \frac{1}{5} * D = \frac{1}{5} * 0.467m = 0.0933m$$

$$Shaft Diameter = \frac{d}{D} * D = \frac{2}{25} * 0.467m = 0.0373m$$

Hydrodynamic Regime Calculations

Power and Impeller Speed Calculations

$$Estimated Power = \frac{P}{V} * V = 0.75 \frac{kW}{m^3} * 2m^3 = 1.5kW$$

Agitation guessing N as 3.83 s^{-1}

$$P = N_p * \rho * N^3 * D^5 = 1.27 * 955.72 \frac{kg}{m^3} * (3.83 \frac{1}{s})^3 * (0.467m)^5 = 1.509kW$$

Power is close to the estimated power, so N can be taken as 3.83 s^{-1}

Reynold's Number

$$Re = \frac{\rho * N * D^2}{\mu} = \frac{(955.72 \frac{kg}{m^3}) * (3.83 \frac{1}{s}) * (0.467m)^2}{0.0015 Pa * s} = 531436 \approx 5.3 * 10^5$$

Froude Number

$$Fr = \frac{N^2 * D}{g} = \frac{(3.83 \frac{1}{s})^2 * (0.467m)}{9.81 \frac{m}{s^2}} = 0.69$$

Mixing Performance, and Energy Calculations

Pumping Capacity

$$Q = N_Q * N * D^3 = 0.79 * 3.83 \frac{1}{s} * (0.467m)^3 = 0.308 \frac{m^3}{s}$$

Mixing Time

$$t_{mix} = k * \frac{T^2}{N * D^3} = 5 * \frac{1.4m^2}{3.83 \frac{1}{s} * (0.467m)^3} = 25.18s$$

Required Energy

$$E = P * t_{mix} = 1.51kW * 25.18s = 38kJ$$

Temperature Change

$$\Delta T = \frac{E}{m_{total} * C_{p(20\%ethanol)}} = \frac{38kJ}{1911.44kg * 3.7 \frac{kJ}{kg * K}} = 0.00537K$$